

Supplementary Material for *On the Minimum Distance of High-rate Moore-Type Quasi-Cyclic LDPC Codes*

Detailed Algebraic Derivations for Weight-8 Cases II–VII

1 Scope and relation to the main manuscript

This supplementary document provides the detailed algebraic derivations omitted from the main manuscript for Cases II–VII in the characterization of Hamming-weight-8 codewords of $\text{Moore}(m, a, b, 3)$ under the assumption $\text{ord}_m(b) = 3$. Case I is derived in full in the main manuscript; the purpose of the present document is to supply the corresponding calculations for the remaining six matching configurations. The notation, normalization, case labels, and condition labels (a)–(d) are those of the main manuscript.

The correspondence between the case labels used below and the matching subcases is

Case	Matching subcase	RREF	Free variables
II	1-2-1	A_{121}	u_4, u_7, u_8
III	1-2-2	A_{122}	u_6, u_7, u_8
IV	2-1-2	A_{212}	u_4, u_6, u_7, u_8
V	2-1-7	A_{217}	u_4, u_6, u_7, u_8
VI	2-3-5	A_{235}	u_4, u_6, u_7, u_8
VII	2-4-3	A_{243}	u_5, u_6, u_7, u_8

2 Algebraic framework and conventions

All calculations are carried out in the finite field $\mathbb{F}_m$. We use the normalization $c_1 = (0, 0)$ and write

$$u_i = a^{t_i}, \quad u_1 = 1.$$

Since $\text{ord}_m(b) = 3$, we have $b \neq 1$ and

$$1 + b + b^2 = 0.$$

All coefficients below are simplified using this identity.

For a matching edge $\{c_i, c_q\}$ at level $s \in \{0, 1, 2\}$, the basic relation is

$$x_i - x_q = (b^s - 1)(u_q - u_i).$$

For an oriented cycle

$$\underline{i_1, i_2, \dots, i_\ell}(s_1 s_2 \cdots s_\ell),$$

the underlined indices denote the successive vertices and s_r is the level of the edge joining c_{i_r} to $c_{i_{r+1}}$, with $i_{\ell+1} = i_1$. Summing the basic edge relations around the cycle eliminates the x -coordinates and gives a homogeneous linear equation in $u_1, \dots, u_8$.

For each matching configuration we proceed as follows. First, we choose a spanning tree and form its fundamental-cycle system. Second, we row-reduce the resulting homogeneous system in the fixed column order $(u_1, \dots, u_8)$; zero rows are omitted. Third, after imposing $u_1 = 1$, the first pivot relation gives condition (b) , while the remaining pivot-variable expressions give condition (d) through membership in $\langle a \rangle$. Finally, the level-0 coordinates and within-block distinctness conditions are reduced, modulo the main congruence, to conditions (a) and (c) . Factors that differ by a nonzero scalar in $\mathbb{F}_m$ are identified, since they define the same nonvanishing condition.

Labeling convention for Case VI. For subcase 2-3-5, we use the representative appearing in the proposition of the main manuscript. Relative to the intermediate labeling arising during the enumeration, this representative is obtained by the relabeling $c_7 \leftrightarrow c_8$. This convention fixes the column order and yields the matrix A_{235} reported in the main manuscript.

3 Case II: Subcase 1-2-1

3.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3, c_6, c_8\}, \{c_2, c_5\}, \{c_4, c_7\}\}, \\ M_2 &= \{\{c_1, c_4\}, \{c_2, c_3\}, \{c_5, c_8\}, \{c_6, c_7\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_1, c_6\}_1, \{c_1, c_8\}_1, \{c_2, c_5\}_1, \{c_4, c_7\}_1, \{c_1, c_4\}_2, \{c_2, c_3\}_2.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below.

- (1) 3, 1, 6(111). The resulting cycle equation is

$$0 = 0.$$

- (2) 3, 1, 8(111). The resulting cycle equation is

$$0 = 0.$$

- (3) 6, 1, 8(111). The resulting cycle equation is

$$0 = 0.$$

- (4) 5, 2, 3, 1, 8(12112). The resulting cycle equation is

$$(1 + 2b)u_2 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_8 = 0.$$

- (5) 6, 1, 4, 7(1212). The resulting cycle equation is

$$(1 + 2b)u_1 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_7 = 0.$$

- (6) 1, 3, 2(120). The resulting cycle equation is

$$(1 - b)u_1 + (-2 - b)u_2 + (1 + 2b)u_3 = 0.$$

(7) 3, 1, 4(120). The resulting cycle equation is

$$(1 + 2b) u_1 + (1 - b) u_3 + (-2 - b) u_4 = 0.$$

(8) 5, 2, 3, 1, 6(12110). The resulting cycle equation is

$$(1 + 2b) u_2 + (-1 - 2b) u_3 + (1 - b) u_5 + (-1 + b) u_6 = 0.$$

(9) 7, 4, 1, 8(1210). The resulting cycle equation is

$$(-1 - 2b) u_1 + (1 + 2b) u_4 + (1 - b) u_7 + (-1 + b) u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{121} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 1+b & -1-b \\ 0 & 1 & 0 & -1 & 0 & 0 & 2+b & -2-b \\ 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1+b & -2-b \\ 0 & 0 & 0 & 0 & 0 & 1 & b & -1-b \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

3.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_4 + (-1 - b) u_7 + (1 + b) u_8, \\ u_2 &= u_4 + (-2 - b) u_7 + (2 + b) u_8, \\ u_3 &= u_4 - u_7 + u_8, \\ u_5 &= (-1 - b) u_7 + (2 + b) u_8, \\ u_6 &= -b u_7 + (1 + b) u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + (-1 - b) u_7 + (1 + b) u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

3.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + (1 + b) u_7 + (-1 - b) u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (1 + 2b) u_7 + (-1 - 2b) u_8, \\ x_5 &= x_6 = -1 + b + (-1 - 2b) u_7 + (2 + b) u_8, \\ x_7 &= x_8 = -1 + b + (1 - b) u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + bu_7 + (-1 - b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + 2bu_7 + (-1 - 2b)u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_8 \neq 0 &\iff 1 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + bu_7 + (-1 - b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + 2bu_7 + (-1 - 2b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

4 Case III: Subcase 1-2-2

4.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4, c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_5\}, \{c_4, c_7\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_4\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_5, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_5\}_1, \{c_4, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_4\}_2, \{c_2, c_6\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 16 edges, it determines 9 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 4, 7(1212). The resulting cycle equation is

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_4 + (1 + 2b)u_7 = 0.$$

(2) 5, 2, 6, 8(1212). The resulting cycle equation is

$$(1 + 2b)u_2 + (-1 - 2b)u_5 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 4(120). The resulting cycle equation is

$$(1 + 2b)u_1 + (1 - b)u_3 + (-2 - b)u_4 = 0.$$

(4) 3, 1, 2, 5(1010). The resulting cycle equation is

$$(-1 + b)u_1 + (1 - b)u_2 + (1 - b)u_3 + (-1 + b)u_5 = 0.$$

(5) 3, 1, 2, 6(1020). The resulting cycle equation is

$$(-1 + b)u_1 + (2 + b)u_2 + (1 - b)u_3 + (-2 - b)u_6 = 0.$$

(6) 4, 1, 2, 5(2010). The resulting cycle equation is

$$(-2 - b)u_1 + (1 - b)u_2 + (2 + b)u_4 + (-1 + b)u_5 = 0.$$

(7) 4, 1, 2, 6(2020). The resulting cycle equation is

$$(-2 - b)u_1 + (2 + b)u_2 + (2 + b)u_4 + (-2 - b)u_6 = 0.$$

(8) 5, 2, 6(120). The resulting cycle equation is

$$(1 + 2b)u_2 + (1 - b)u_5 + (-2 - b)u_6 = 0.$$

(9) 7, 4, 1, 2, 6, 8(120210). The resulting cycle equation is

$$\begin{aligned} &(-2 - b)u_1 + (2 + b)u_2 + (1 + 2b)u_4 + (-1 - 2b)u_6 \\ &+ (1 - b)u_7 + (-1 + b)u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{122} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 + b & -1 & 1 - b \\ 0 & 1 & 0 & 0 & 0 & -1 + b & 0 & -b \\ 0 & 0 & 1 & 0 & 0 & b & -1 & -b \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & b & 0 & -1 - b \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

4.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= (1 - b)u_6 + u_7 + (-1 + b)u_8, \\ u_2 &= (1 - b)u_6 + bu_8, \\ u_3 &= -bu_6 + u_7 + bu_8, \\ u_4 &= u_6 + u_7 - u_8, \\ u_5 &= -bu_6 + (1 + b)u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + (1 - b)u_6 + u_7 + (-1 + b)u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

4.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_7 :

$$u_7 = 1 + (-1 + b)u_6 + (1 - b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = x_5 = x_6 = (-1 + b)u_6 + (1 - b)u_8, \\ x_7 &= x_8 = (-2 + 2b)u_6 + (2 - 2b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 + b)u_6 - bu_8 &\not\equiv 0 \pmod{m}, \\ 1 - 2u_6 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + 2bu_6 + (-1 - 2b)u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_6 &\neq t_8 \iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_7 &\neq t_8 \iff 1 + (-1 + b)u_6 - bu_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 - 2u_6 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + 2bu_6 + (-1 - 2b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

5 Case IV: Subcase 2-1-2

5.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_3, c_6\}, \{c_2, c_7\}, \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_4\}_1, \{c_5, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_7\}_2, \{c_3, c_6\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 4, 2, 7, 5, 1, 3, 6, 8(12121212). The resulting cycle equation is

$$\begin{aligned} &(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_4 \\ &+ (1 + 2b)u_5 + (-1 - 2b)u_6 + (-1 - 2b)u_7 + (1 + 2b)u_8 = 0. \end{aligned}$$

(2) 1, 5, 7, 2(2120). The resulting cycle equation is

$$(2 + b) u_1 + (-2 - b) u_2 + (-1 - 2b) u_5 + (1 + 2b) u_7 = 0.$$

(3) 3, 1, 5, 7, 2, 4(121210). The resulting cycle equation is

$$(1 + 2b) u_1 + (-1 - 2b) u_2 + (1 - b) u_3 + (-1 + b) u_4 \\ + (-1 - 2b) u_5 + (1 + 2b) u_7 = 0.$$

(4) 5, 1, 3, 6(2120). The resulting cycle equation is

$$(-1 - 2b) u_1 + (1 + 2b) u_3 + (2 + b) u_5 + (-2 - b) u_6 = 0.$$

(5) 7, 5, 1, 3, 6, 8(121210). The resulting cycle equation is

$$(-1 - 2b) u_1 + (1 + 2b) u_3 + (1 + 2b) u_5 + (-1 - 2b) u_6 \\ + (1 - b) u_7 + (-1 + b) u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{212} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -1 - b & b & 1 \\ 0 & 1 & 0 & -1 & 0 & 0 & b & -b \\ 0 & 0 & 1 & -1 & 0 & -1 - b & 0 & 1 + b \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

5.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$u_1 = u_4 + (1 + b) u_6 - bu_7 - u_8, \\ u_2 = u_4 - bu_7 + bu_8, \\ u_3 = u_4 + (1 + b) u_6 + (-1 - b) u_8, \\ u_5 = u_6 + u_7 - u_8.$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + (1 + b) u_6 - bu_7 - u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

5.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + (-1 - b)u_6 + bu_7 + u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (1 + 2b)u_7 + (-1 - 2b)u_8, \\ x_5 &= x_6 = -2 - b + (2 + b)u_6 + (2 + b)u_7 + (-2 - b)u_8, \\ x_7 &= x_8 = -2 - b + (1 + 2b)u_6 + (2 + b)u_7 + (-1 - 2b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 - u_7 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 + bu_7 - bu_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 + bu_7 &\not\equiv 0 \pmod{m}, \\ u_6 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 - u_6 - u_7 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c3} &\iff 1 - u_6 + bu_7 - bu_8 \not\equiv 0 \pmod{m}, \\ \text{c4} &\iff 1 + (-1 - b)u_6 + bu_7 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

6 Case V: Subcase 2-1-7

6.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_4\}_1, \{c_5, c_7\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_6\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 5, 7(1212). The resulting cycle equation is

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_7 = 0.$$

(2) 4, 2, 6, 8(1212). The resulting cycle equation is

$$(1 + 2b)u_2 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 2, 4(1010). The resulting cycle equation is

$$(-1 + b)u_1 + (1 - b)u_2 + (1 - b)u_3 + (-1 + b)u_4 = 0.$$

(4) 5, 1, 2, 6(2020). The resulting cycle equation is

$$(-2 - b)u_1 + (2 + b)u_2 + (2 + b)u_5 + (-2 - b)u_6 = 0.$$

(5) 7, 5, 1, 2, 6, 8(120210). The resulting cycle equation is

$$\begin{aligned} &(-2 - b)u_1 + (2 + b)u_2 + (1 + 2b)u_5 + (-1 - 2b)u_6 \\ &+ (1 - b)u_7 + (-1 + b)u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{217} = \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & -1 & -1 & 2 \\ 0 & 1 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

6.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$u_1 = u_4 + u_6 + u_7 - 2u_8,$$

$$u_2 = u_4 + u_6 - u_8,$$

$$u_3 = u_4 + u_7 - u_8,$$

$$u_5 = u_6 + u_7 - u_8.$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_4 + u_6 + u_7 - 2u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

6.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_7 :

$$u_7 = 1 - u_4 - u_6 + 2u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (-1 + b)u_6 + (1 - b)u_8, \\ x_5 &= x_6 = (-2 - b)u_4 + (2 + b)u_8, \\ x_7 &= x_8 = (-2 - b)u_4 + (-1 + b)u_6 + 3u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ u_4 - u_8 &\not\equiv 0 \pmod{m}, \\ u_4 - bu_6 + (-1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ u_4 + bu_6 + (-1 - b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - u_4 - u_6 + u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_4 \neq t_8 &\iff u_4 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff 1 - u_4 - u_6 + u_8 \not\equiv 0 \pmod{m}, \\ c1 &\iff u_4 + bu_6 + (-1 - b)u_8 \not\equiv 0 \pmod{m}, \\ c2 &\iff u_4 - bu_6 + (-1 + b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

7 Case VI: Subcase 2-3-5

7.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_6\}, \{c_5, c_7\}, \{c_4, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_2, c_4\}, \{c_3, c_7\}, \{c_6, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_6\}_1, \{c_4, c_8\}_1, \{c_5, c_7\}_1, \{c_1, c_5\}_2, \{c_2, c_4\}_2, \{c_1, c_2\}_0.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 3, 1, 5, 7(1212). The resulting cycle equation is

$$(1 + 2b)u_1 + (-1 - 2b)u_3 + (-1 - 2b)u_5 + (1 + 2b)u_7 = 0.$$

(2) 6, 2, 4, 8(1212). The resulting cycle equation is

$$(1 + 2b)u_2 + (-1 - 2b)u_4 + (-1 - 2b)u_6 + (1 + 2b)u_8 = 0.$$

(3) 3, 1, 2, 4(1020). The resulting cycle equation is

$$(-1 + b)u_1 + (2 + b)u_2 + (1 - b)u_3 + (-2 - b)u_4 = 0.$$

(4) 5, 1, 2, 6(2010). The resulting cycle equation is

$$(-2 - b)u_1 + (1 - b)u_2 + (2 + b)u_5 + (-1 + b)u_6 = 0.$$

(5) 7, 5, 1, 2, 4, 8(120210). The resulting cycle equation is

$$\begin{aligned} &(-2 - b)u_1 + (2 + b)u_2 + (-1 - 2b)u_4 + (1 + 2b)u_5 \\ &+ (1 - b)u_7 + (-1 + b)u_8 = 0. \end{aligned}$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{235} = \begin{bmatrix} 1 & 0 & 0 & b & 0 & -1 - b & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & b & 0 & 0 & -1 & -b \\ 0 & 0 & 0 & 0 & 1 & -1 - b & -1 & 1 + b \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

7.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_4, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= -bu_4 + (1 + b)u_6 + u_7 - u_8, \\ u_2 &= u_4 + u_6 - u_8, \\ u_3 &= -bu_4 + u_7 + bu_8, \\ u_5 &= (1 + b)u_6 + u_7 + (-1 - b)u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 - bu_4 + (1 + b)u_6 + u_7 - u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d).

7.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_4 :

$$u_4 = 1 + b - bu_6 + (-1 - b)u_7 + (1 + b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = (-2 - b)u_6 + (2 + b)u_8, \\ x_5 &= x_6 = -2 - b + (1 + 2b)u_6 + (2 + b)u_7 + (-1 - 2b)u_8, \\ x_7 &= x_8 = -2 - b + (-1 + b)u_6 + (2 + b)u_7 + (1 - b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} u_6 - u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 - bu_6 - u_7 + bu_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-2 - b)u_6 - u_7 + (2 + b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-2 - b)u_6 + bu_7 + u_8 &\not\equiv 0 \pmod{m}, \\ 1 + (-1 - b)u_6 + bu_7 &\not\equiv 0 \pmod{m}, \\ u_6 + (-1 - b)u_7 + bu_8 &\not\equiv 0 \pmod{m}, \\ u_7 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_4 \neq t_8 &\iff 1 + (-1 - b)u_6 - u_7 + (1 + b)u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + (-2 - b)u_6 + bu_7 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff 1 + (-1 - b)u_6 + bu_7 \not\equiv 0 \pmod{m}, \\ \text{c3} &\iff u_6 + (-1 - b)u_7 + bu_8 \not\equiv 0 \pmod{m}, \\ \text{c4} &\iff 1 + (-2 - b)u_6 - u_7 + (2 + b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c5} &\iff 1 - bu_6 - u_7 + bu_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

8 Case VII: Subcase 2-4-3

8.1 Matching configuration and fundamental-cycle system

The matching set is

$$\begin{aligned} M_0 &= \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 &= \{\{c_1, c_3\}, \{c_2, c_7\}, \{c_4, c_5\}, \{c_6, c_8\}\}, \\ M_2 &= \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}. \end{aligned}$$

We choose the spanning tree

$$F = \{c_1, c_3\}_1, \{c_2, c_7\}_1, \{c_4, c_5\}_1, \{c_6, c_8\}_1, \{c_1, c_5\}_2, \{c_2, c_6\}_2, \{c_3, c_7\}_2.$$

This tree has 7 edges. Since the matching graph has 12 edges, it determines 5 fundamental cycles. The associated cycle equations are listed below.

(1) 4, 5, 1, 3, 7, 2, 6, 8(12121212). The resulting cycle equation is

$$(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_4 \\ + (1 + 2b)u_5 + (-1 - 2b)u_6 + (-1 - 2b)u_7 + (1 + 2b)u_8 = 0.$$

(2) 1, 3, 7, 2(1210). The resulting cycle equation is

$$(1 - b)u_1 + (-1 + b)u_2 + (1 + 2b)u_3 + (-1 - 2b)u_7 = 0.$$

(3) 3, 1, 5, 4(1210). The resulting cycle equation is

$$(1 + 2b)u_1 + (1 - b)u_3 + (-1 + b)u_4 + (-1 - 2b)u_5 = 0.$$

(4) 5, 1, 3, 7, 2, 6(212120). The resulting cycle equation is

$$(-1 - 2b)u_1 + (1 + 2b)u_2 + (1 + 2b)u_3 + (2 + b)u_5 \\ + (-2 - b)u_6 + (-1 - 2b)u_7 = 0.$$

(5) 7, 2, 6, 8(1210). The resulting cycle equation is

$$(1 + 2b)u_2 + (-1 - 2b)u_6 + (1 - b)u_7 + (-1 + b)u_8 = 0.$$

After deleting zero equations and row-reducing the remaining system in the column order $(u_1, \dots, u_8)$, we obtain

$$A_{243} = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & -1 - b & 1 + b \\ 0 & 1 & 0 & 0 & 0 & -1 & -1 - b & 1 + b \\ 0 & 0 & 1 & 0 & -1 - b & 1 + b & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 - b & 1 + b & 0 & -1 \end{bmatrix}.$$

Thus the fundamental-cycle system has the RREF reported in the main manuscript.

8.2 RREF solution and conditions (b) and (d)

The free variables are

$$u_5, u_6, u_7, u_8.$$

The pivot variables are therefore

$$\begin{aligned} u_1 &= u_5 + (1 + b)u_7 + (-1 - b)u_8, \\ u_2 &= u_6 + (1 + b)u_7 + (-1 - b)u_8, \\ u_3 &= (1 + b)u_5 + (-1 - b)u_6 + u_7, \\ u_4 &= (1 + b)u_5 + (-1 - b)u_6 + u_8. \end{aligned}$$

Imposing $u_1 = 1$ gives the main congruence

$$-1 + u_5 + (1 + b)u_7 + (-1 - b)u_8 = 0,$$

which is condition (b) after substituting $u_i = a^{t_i}$. Requiring the remaining dependent variables above to belong to $\langle a \rangle$ gives condition (d) .

8.3 Distinctness conditions (a) and (c)

For the distinctness calculation we solve the main congruence for u_5 :

$$u_5 = 1 + (-1 - b)u_7 + (1 + b)u_8.$$

After this substitution, the common x -coordinate of each block of M_0 is

$$\begin{aligned} x_1 &= x_2 = 0, \\ x_3 &= x_4 = 1 + 2b + (-2 - b)u_6 - 3bu_7 + (1 + 2b)u_8, \\ x_5 &= x_6 = (-1 - 2b)u_7 + (1 + 2b)u_8, \\ x_7 &= x_8 = (-1 + b)u_6 + (-1 - 2b)u_7 + (2 + b)u_8. \end{aligned}$$

Hence different level-0 blocks have distinct x -coordinates, and the columns inside each block are distinct, exactly when the following factors are all nonzero (factors differing by a nonzero scalar are identified):

$$\begin{aligned} 1 + bu_6 + (-2 - b)u_7 + u_8 &\not\equiv 0 \pmod{m}, \\ u_7 - u_8 &\not\equiv 0 \pmod{m}, \\ u_6 + bu_7 + (-1 - b)u_8 &\not\equiv 0 \pmod{m}, \\ 1 + bu_6 + (-1 - b)u_7 &\not\equiv 0 \pmod{m}, \\ 1 - u_6 + (-1 - b)u_7 + (1 + b)u_8 &\not\equiv 0 \pmod{m}, \\ u_6 - u_8 &\not\equiv 0 \pmod{m}. \end{aligned}$$

Using the main congruence, these factors reduce to condition (a) together with the nonzero factors in condition (c). Explicitly,

$$\begin{aligned} \text{a1: } t_5 \neq t_6 &\iff 1 - u_6 + (-1 - b)u_7 + (1 + b)u_8 \not\equiv 0 \pmod{m}, \\ \text{a2: } t_6 \neq t_8 &\iff u_6 - u_8 \not\equiv 0 \pmod{m}, \\ \text{a3: } t_7 \neq t_8 &\iff u_7 - u_8 \not\equiv 0 \pmod{m}, \\ \text{c1} &\iff 1 + bu_6 + (-2 - b)u_7 + u_8 \not\equiv 0 \pmod{m}, \\ \text{c2} &\iff u_6 + bu_7 + (-1 - b)u_8 \not\equiv 0 \pmod{m}, \\ \text{c3} &\iff 1 + bu_6 + (-1 - b)u_7 \not\equiv 0 \pmod{m}, \\ \text{c4} &\iff 1 - u_6 + (-1 - b)u_7 + (1 + b)u_8 \not\equiv 0 \pmod{m}. \end{aligned}$$

Hence conditions (a) and (c) are precisely the level-0 and within-block distinctness requirements for this matching configuration. Together with conditions (b) and (d), they give the necessary and sufficient algebraic conditions for this case.

9 Conclusion

For each of Cases II–VII, the fundamental-cycle equations derived from the corresponding matching configuration reduce to the RREF stated in the main manuscript. Under the normalization $u_1 = 1$, the pivot relations yield condition (b) and the subgroup-membership requirements in condition (d). The remaining level-0 and within-block distinctness constraints yield exactly conditions (a) and (c). Consequently, the calculations in this supplement complete the algebraic verification of the six alternatives omitted from the detailed proof of the Hamming-weight-8 existence theorem.